

Jainam Shah

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EDUCATION

New Jersey Institute of Technology | Albert Dorman Honors College

GPA: 3.94

B.S. in Computer Science | Minors in Business, Math, and AI

Sep. 2023 – May 2027

EXPERIENCE

Capital One | Software Engineering Intern

Jun. 2026 – Present

- Building a **robust LLM** platform by spawning **parallel Claude** agents to iteratively refine **product owner** ideas into engineer-ready **PRDs**, **tech specs**, and **architecture diagrams**, cutting the spec-validation process by **90%**
- Developing an **SSE pipeline** streaming **tokenized AI output** to sustain **55s+ generations** without timeouts
- Designing **S3-triggered Lambda** pipelines to persist **Claude** context to **DynamoDB** for conversation resumption
- Enforce **least-privilege RBAC** across **50+** accounts, restricting access to sensitive product roadmaps and specs

Prudential Financial | Software Engineering Intern

Sep. 2025 – Dec. 2025

- Created an **AWS Bedrock** chatbot with **Lambda** and **boto3** to create **EKS**, **EC2**, and **S3** resources on demand
- Enabled Bedrock agents to manage clusters for **100+ teams** by building an **MCP server** via **Kubernetes API**
- Deployed AWS Bedrock agent and action groups via **Terraform**, allowing access to **50+ AWS environments**
- Fixed **Azure** Kubernetes auth by replacing **kubelogin** with a custom **IAM flow** for unified **AWS MCP** access

TD Securities | Software Engineering Intern

Jun. 2025 – Aug. 2025

- Leveraged **Java Spring Boot** to monitor **80+** APIs over **REST** and **ZeroMQ** through a **real-time** dashboard
- Cut execution time by **70%** by **parallelizing** API checks and trade metric validation via **multithreaded pipeline**
- Employed **gRPC** to drop latency by **52%** for **4+** backend services, accelerating **cross-platform** communication
- Streamed **API** payloads to detect inconsistencies between live valuations and known data across **1,000+ trades**

Big Data Analytics Laboratory | Machine Learning Intern

Feb. 2024 – Dec. 2024

- Applied **machine learning** to help the **U.S. Navy** predict ship maintenance delays and optimize resource allocation
- Enhanced model accuracy by **80%**, reducing predictive errors to **14 days** and saving the U.S. Navy over **\$1 billion**
- Analyzed over **15k** features, yielding data-driven recommendations on model functionality for **U.S. Navy** officials
- Reduced overfitting by **20%** and increased **accuracy** by **10%** by applying a custom **ML harmonization** algorithm

Tech For Good Inc. | Software Engineering Intern

Jul. 2024 – Sep. 2024

- Engineered an **openEHR-compliant** EMR system in **Go** to digitize rural patient records at Tanzanian hospitals
- Implemented a **ScyllaDB** schema with **5000+** records, exposed via performant **GraphQL API** endpoints
- Employed Docker to ensure consistent database and language schemas across local machines for all team members

PROJECTS

SentiStock | *Flask, Hugging Face (FinBERT and BART), SerpAPI, DiffBot*

- Built an **AI-powered** stock sentiment analyzer using **FinBERT** & **BART** for classification and summarization
- Integrated **SerpAPI** and **DiffBot** for news analysis, delivering transparent buy/hold/sell insights via a web UI

JobHub | *React, FastAPI, Supabase, Groq, Vercel*

- Built a **React** and **FastAPI** ATS using **Supabase** Postgres, **Auth**, and **Storage** for jobs, documents, and profiles
- Integrated **Groq** LLMs for AI cover-letter drafts, resume rewrites, and per-job company research via **FastAPI**
- Developed document versioning, job analytics, reminders, and interview tracking with accessible **React** dashboards

PayPark | *(Columbia DivHacks '25: 3rd Overall, Best Financial Hack (Capital One) & Best Use of XRP Ledger)*

- Engineered a **Next.js**, **Express**, **FastAPI** & **Prisma** stack for a seamless UI and efficient backend operations
- Integrated **XRPLedger** for automated parking payments, wallet syncing, and real-time transaction tracking
- Utilized **TomTom API**, **WebSocket** services, and **Echo AI** for parking pricing and session management

TECHNICAL SKILLS

Languages: Bash, C, C++, Golang, HTML/CSS, Matlab, Java, JavaScript, Python, PHP, SQL, TypeScript

Frameworks: Django, Express.js, FastAPI, Flask, gRPC, Java Spring Boot, JUnit, Node.js, Next.js, Prisma, React

Developer Tools: AWS, Docker, Git, GraphQL, Jira, Kubernetes, MCP, Postman, REST, Supabase, Terraform, ZeroMQ

Machine Learning: Hugging Face, Jupyter Notebooks, NumPy, Pandas, PyTorch, Scikit-learn, TensorFlow